

Lesson 44 — SLAM: How a Robot Builds Its Own Map

Module 18 — SLAM & Autonomous Navigation

The robot can see, track, and measure distance. Now it has to answer a harder pair of questions at the same time: **where am I, and what does this place look like?**

We already met the core idea in [Lesson 27](#) with an ASCII occupancy grid. This lesson is the industry version: LiDAR SLAM vs visual SLAM, odometry drift, loop closure, ROS 2, and a **real HD video** you can post on LinkedIn.

Lesson Goal

- What SLAM, localization, and mapping each mean — and why they are not the same thing
- Why indoor robots cannot rely on GPS
- LiDAR SLAM, visual SLAM, and visual-inertial SLAM
- Odometry, drift, and loop closure
- The names you will see in ROS 2: SLAM Toolbox, Cartographer, RTAB-Map, ORB-SLAM3
- A visual-odometry overlay on a free Full-HD clip

1. What SLAM Is

SLAM = Simultaneous Localization and Mapping.

You leave a robot in a house it has never seen. Nobody gives it a floor plan.

```
Robot
↓
looks at the room
↓
detects walls
↓
starts a map
↓
estimates its own position on that map
↓
moves
↓
updates the map
```

It is building the map **and** locating itself on that map at the same time. That circular dependency is the whole difficulty.

2. Mapping

Mapping asks: **what does this place look like?**



Walls, obstacles, and free space become cells (2D occupancy grid) or points (3D point cloud). A map without a pose is a picture you cannot use.

3. Localization

Localization asks: **where am I on the map?**



A pose without a map is a coordinate in an empty universe. The robot needs both.

4. SLAM Combines Them



A fourth word sits on top: **navigation** — "where do I go next?" That is Lesson 45 / Nav2. Do not mix the four up:

Term	Question
Mapping	What does the environment look like?
Localization	Where am I?
Navigation	Where do I need to go?
SLAM	Build the map and estimate the pose, together

5. Why Not Just GPS?

Outdoors, GPS is useful to a few metres. Indoors:

 walls block the satellites

A robot vacuum, a warehouse AMR, a hospital trolley — none of them get a trustworthy GPS fix. They use **LiDAR, cameras, IMUs, and wheel encoders** instead.

6. LiDAR SLAM

LiDAR fires laser beams and times the reflection.



Each beam is a range. A full scan is a slice of the room. Stack scans as the robot moves and you get a geometric map — the occupancy grid from Lesson 27.

Hallway:



LiDAR sees the left wall, the right wall, whatever is in front. The map grows scan by scan.

7. Visual SLAM

Same job, a camera instead of a laser.

```
Camera
↓
image
↓
features (corners, edges)
↓
motion estimate
↓
map + pose
```

The robot does not need to "understand" the room. It needs **features it can recognise again** — book spines, door frames, the corner of a table.

8. LiDAR vs Visual

LiDAR SLAM	Visual SLAM
Laser ranges	Camera images
Accurate geometry	Rich appearance
Expensive sensor	A camera is cheap
Works in the dark	Needs light (or an IR camera)
Strong at mapping walls	Strong at recognising places

Both are used in production. Robot vacuums are usually LiDAR (or a mix). Tesla's stack is famously camera-first. Warehouse robots often run LiDAR + wheel odometry, with cameras for extras.

9. Visual-Inertial SLAM

```
Camera + IMU → Visual-Inertial SLAM
```

The IMU measures acceleration and angular velocity. Between camera frames it tells you how the robot rotated and lurched. The camera corrects the IMU's drift; the IMU covers the blurry frames. Phones (ARKit, ARCore) and drones do this. ORB-SLAM3 can run in this mode.

10. Odometry — and Why It Drifts

Odometry: **estimate motion from the robot's own movement sensors.**

```
wheel encoder → rotation → distance travelled → pose
```

"I have moved about one metre since the last tick."

The problem is **drift**. Every small slip, rounding error, and unmodelled bump is added to the next one. After a long corridor the believed pose and the true pose have quietly walked apart:

```
actual      🤖  
estimated   🤖    ← same robot, two answers
```

Lesson 27's second pass is exactly this: correct laser hits written into the wrong cells, walls doubling up, accuracy collapsing.

🔄 11. Loop Closure

SLAM's answer to drift: **recognise a place you have already been.**

```
START  
↓  
A → B → C  
↑       ↓  
F ← E ← D  
↓  
back at START → "I know this doorway"  
↓  
correct the whole trajectory  
↓  
the map snaps consistent
```

That recognition-and-correction step is **loop closure**. Without it, a twenty-minute mapping run becomes the smeared map from Lesson 27. With it, the robot can map a whole floor.

🏗️ 12. The Pipeline

```
Sensors  
↓  
Odometry  
↓  
Scan / feature matching  
↓  
Pose estimation  
↓  
Map update  
↓  
Loop closure  
↓  
Optimised map
```

Remember this architecture. Every SLAM package is a different implementation of the same diagram.

13. Names You Will See

Start recognising these. You do not install them today.

System	What it is
ORB-SLAM3	Research-grade visual / visual-inertial SLAM
RTAB-Map	RGB-D, stereo, and LiDAR; common on real robots
Cartographer	Google's LiDAR SLAM (2D and 3D)
SLAM Toolbox	The default 2D SLAM package in ROS 2

14. ROS 2 Sketch

Later the topics look like this:

```
LiDAR
  ↓
 /scan
  ↓
SLAM Toolbox
  ↓
 /map
  ↓
localization
  ↓
Nav2
  ↓
the robot actually moves
```

That is the first time ROS 2 topics stop being a metaphor and start being the wiring of a real system. We stay on the MacBook until the wiring is clear.

15. Two Real Systems

Robot vacuum

```
start → scan the room → walls → move → update map → next room → finished map
```

Then it remembers kitchen / living room / hall and plans an efficient clean.

Warehouse robot

```
LiDAR → SLAM → warehouse map → localization → Nav2 → Shelf A → packing station
```

Same pipeline, bigger building, tighter tolerances.

The future Jetson stack you have sketched (Orin Nano + LiDAR + depth camera + ROS 2) is this diagram on real hardware. We practise it in simulation first. No purchase required.

Practical — Visual SLAM on a Free HD Clip

You have no LiDAR on the desk. You have a camera stack from Lessons 30–43. That is enough to **watch visual SLAM happen on a real moving camera**.

```
source .venv/bin/activate

python visual_slam.py --test          # the maths, no video
python visual_slam.py --linkedin      # downloads a free Full-HD clip, overlays SLAM
python visual_slam.py --linkedin --reel --gif
```

[visual_slam.py](#) does four things:

1. Downloads Mixkit clip **#21589** (library corridor, 1920×1080, Mixkit Stock Video Free License — commercial use allowed, no faces). Saved to `video_out/` so it is not committed.
2. Tracks hundreds of corners with optical flow — the features a visual-SLAM front-end actually uses.
3. Integrates that motion into a pose (**odometry**) and draws a bird's-eye **map**.
4. Renders a 1920×1080 dashboard: camera + map + the four terms on screen, with a 2-second hook and a 3-second closer. Output: `video_out/slam_linkedin_landscape.mp4` .

Pass `--video your.mp4` to run the same overlay on any clip. A phone pan around your own room is the most honest version.

The notebook [lesson 44 slam.ipynb](#) walks through the same ideas cell by cell.

This demo is **visual odometry**, the front-end of visual SLAM. A production system (ORB-SLAM3, RTAB-Map) also bundle-adjusts the map and closes loops across a full graph. The LinkedIn video is the intuition; those packages are the engineering.

Verified `--test` output:

```
Lesson 44 tests passed.
odometry integrates, loop-closure needs BOTH distance and recognition,
and optical flow tracks a moving texture.
```

Software-First Path

```
MacBook
↓
this demo / Webots / Gazebo
```

```
↓  
virtual LiDAR or a camera  
↓  
SLAM  
↓  
a map
```

Hardware later. Concepts now.

Mini Quiz

	Question	Answer
Q1	SLAM in full?	Simultaneous Localization and Mapping
Q2	What does LiDAR measure?	Distance, by timing a laser
Q3	Estimating the robot's pose?	Localization
Q4	Recognising a previous place and correcting the map?	Loop closure
Q5	Estimating motion from wheel / camera motion?	Odometry

RoboticsFemme — Reel / LinkedIn

Hook (on screen, works muted):

How does a robot build a map when nobody gives it one?

Then the dashboard: features lighting up, the map growing, the pose ticking.

Closer:

The robot is simultaneously learning the world and finding itself inside it.

Shot list and caption: [Reel 7 in docs/reel-scripts.md](#) .

Portfolio

Autonomous robot mapping with SLAM — a LinkedIn-ready overlay on a licensed HD clip, plus the Lesson 27 occupancy-grid demo that proves why mapping without localization is useless.

Next implementation (Lesson 45): the same ideas on a Webots robot with a virtual LiDAR, then ROS 2.

Progress

Lesson **44** / ~250.

Phase	Status
Robotics foundation	done
Basic computer vision	done
AI vision	almost done
SLAM & autonomous navigation	starting now
ROS 2 (practical stack)	next
Jetson / LiDAR hardware	later
Physical AI / humanoids	advanced

Next — Lesson 45: First SLAM Robot in Webots

Theory is in place. Next we put a virtual LiDAR on a robot in Webots, drive it, watch the scans, and grow a map — still with no hardware.

1. Virtual robot
2. Virtual LiDAR
3. Drive and inspect `/scan` -style data
4. Map generation
5. Then ROS 2 wiring